

Multidimensional difference arrays

via Discrete Calculus

What is this talk about?

- A runtime optimization for applying L update queries to an array of size N :

$$\mathcal{O}(L \cdot N) \longrightarrow \mathcal{O}(L + N)$$

- Simple heuristic for 1D, 2D and 3D arrays
- For general dimension: *Discrete Calculus*
- Further optimizations

One-dimensional difference arrays

	0	1	2	3	4	5	6	7	8
A:	*	*	*	*	*	*	*	*	*

↓ Apply query

	0	1	2	3	4	5	6	7	8
A:	*	+C	+C	+C	*	*	*	*	*

General idea for an array of size N and L queries:

"Mark" L queries in $\mathcal{O}(L)$ and perform the updates later in one sweep in $\mathcal{O}(N)$.

	0	1	2	3	4	5	6	7	8
d:	0	0	0	0	0	0	0	0	0

↓ Mark query

	0	1	2	3	4	5	6	7	8
d:	0	+C	0	0	-C	0	0	0	0

↓ Cumulative sum

	0	1	2	3	4	5	6	7	8
Σd :	0	C	C	C	0	0	0	0	0

↓ Cumulative Sum

	0	1	2	3	4	5	6	7	8
A + Σd :	*	+C	+C	+C	*	*	*	*	*

One-dimensional difference arrays

For an array of size N and L queries of arbitrary size

Naive algorithm:

1. For every query add its update value over the whole update range

Runtime: $\mathcal{O}(L \cdot N)$

Non-formal difference array algorithm:

1. Initialize array d of size N with zero entries in $\mathcal{O}(N)$
2. For every query add two markers to d in $\mathcal{O}(L)$
3. Form the cumulative sum Σd in $\mathcal{O}(N)$
4. Add Σd to A in $\mathcal{O}(N)$

Runtime: $\mathcal{O}(L + N)$

One-dimensional difference arrays

Imagine placing the markers as forming separate difference arrays $d_1, \dots, d_L$, that is

$$d = \sum_{i=1}^L d_i$$

The difference array algorithm works as the cumulative sum distributes over the markers of multiple queries:

$$\Sigma d[j] = \sum_{l=1}^j d[l] = \sum_{l=1}^j \sum_{i=1}^L d_i[l] = \sum_{i=1}^L \sum_{l=1}^j d_i[l] = \sum_{i=1}^L \Sigma d_i$$

The cumulative sum Σd can be computed in $\mathcal{O}(N)$ by reusing previous results:

$$\Sigma d[j] = d[j] + \sum_{i=1}^{j-1} d[i] = d[j] + \Sigma d[j-1].$$

Two-dimensional difference arrays

This concept can easily be generalized to 2D arrays.

*	*	*	*	*	*	*	*
*	$+C$	$+C$	$+C$	$+C$	$+C$	*	*
*	$+C$	$+C$	$+C$	$+C$	$+C$	*	*
*	$+C$	$+C$	$+C$	$+C$	$+C$	*	*
*	*	*	*	*	*	*	*

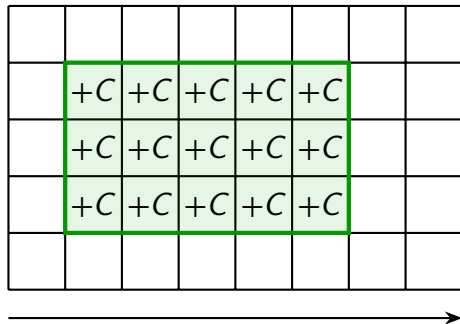
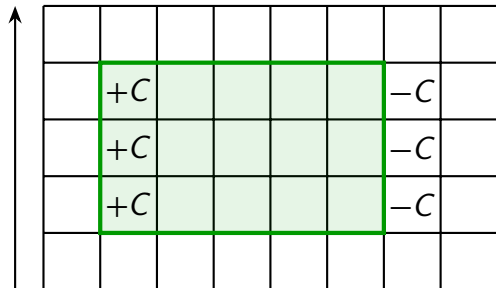
Naively applying the constant update to a 5×8 array A.

	$-C$					$+C$	

Placing *four* markers in the difference array.

Two-dimensional difference arrays

Taking cumulative sums in both directions yields the correct updates.



d -dimensional difference arrays?

- This heuristic can easily be extended to d -dimensional arrays by placing 2^d markers for every query.
- However for dimensions $d \geq 4$ this is hard to conceptually verify.
- Furthermore, "naive" proofs without any additional tools are bound to get very messy.
- This is where *Discrete Calculus* comes into play.
- To leverage these tools, we first need to formally describe our situation.

The formal framework

We view arrays as maps over *finite grid domains*.

Definition (discrete maps, arrays):

- For $a \leq b \in \mathbb{Z}$ define the *discrete interval* $\llbracket a, b \rrbracket := \{m \in \mathbb{Z} \mid a \leq m \leq b\}$
- A *finite grid domain* is a subset $D = \prod_{i=1}^d \llbracket n_i, m_i \rrbracket \subseteq \mathbb{Z}^d$
- A *discrete map* is a map $f : D \rightarrow \mathbb{R}$ over a finite grid domain $D \subseteq \mathbb{Z}^d$
- A *d-dimensional array* is a discrete map $f : \prod_{i=1}^d \llbracket 0, m_i \rrbracket \rightarrow \mathbb{R}$

Definition (constant update queries):

- A discrete map $q : D \rightarrow \mathbb{R}$ which is *constant* on a subset $D_q = \prod_{i=1}^d \llbracket a_i, b_i \rrbracket \subseteq D$ is called *constant update query*
- In other words, $q = C \cdot \mathbb{1}_{D_q}$ with $C \in \mathbb{R}$ (the *update value* of q)
- $|D_q|$ is the *size* of q

Let $f : D \rightarrow \mathbb{R}$ with $D = \prod_{i=1}^d \llbracket n_i, m_i \rrbracket$ be a discrete function and $q_1, \dots, q_L$ a series of constant update queries.

Applying the series of queries to f means forming the updated discrete function $F : D \rightarrow \mathbb{R}$ as

$$F := f + \sum_{i=1}^L q_i.$$

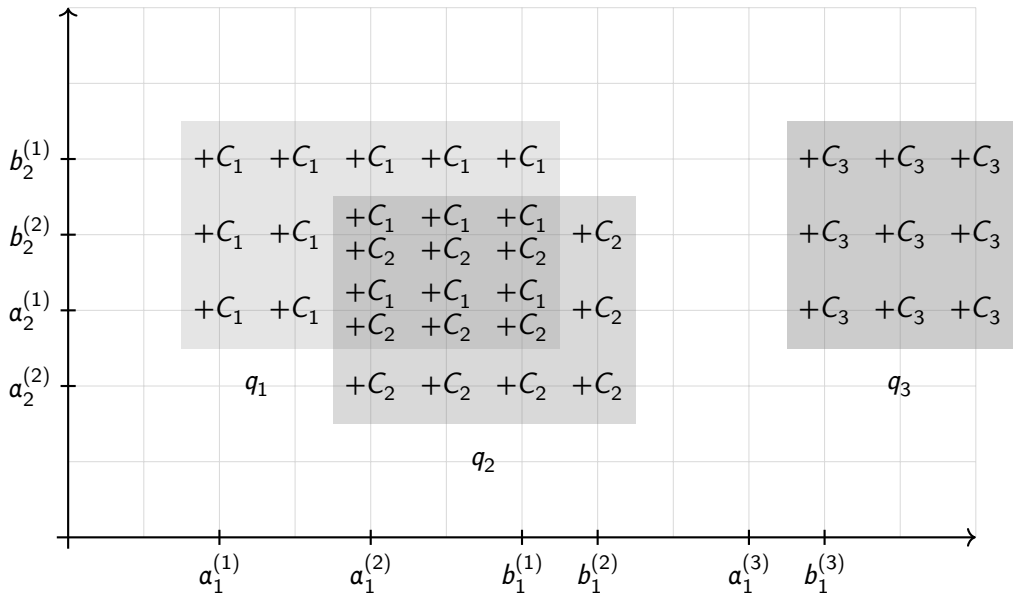
A naive algorithm

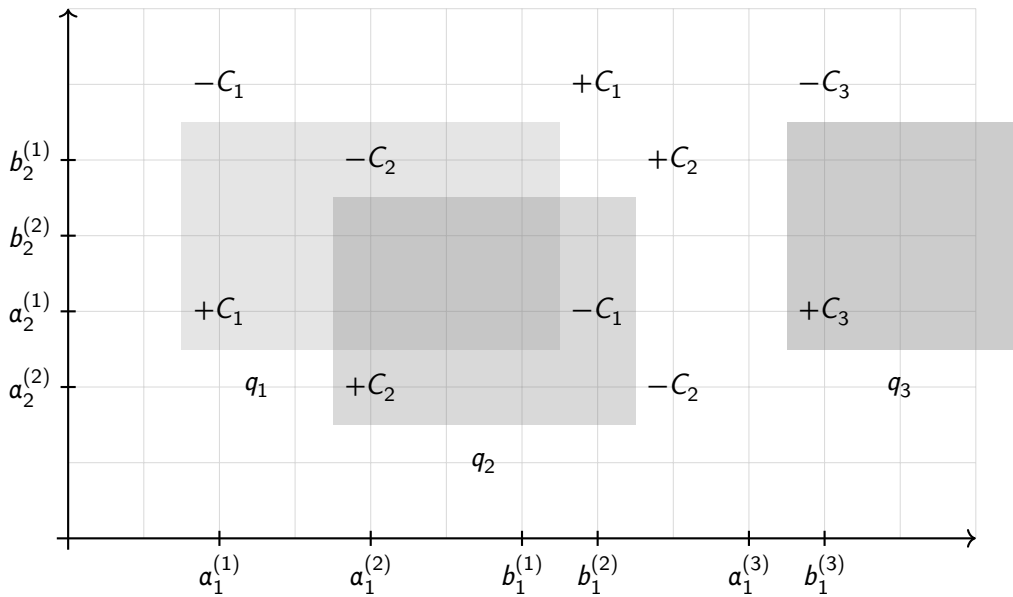
Input: A discrete map $f : D \rightarrow \mathbb{R}$ with $D = \prod_{i=1}^d \llbracket n_i, m_i \rrbracket$ and a series of constant update queries $q_1, \dots, q_L$ on f over the update ranges $D_j = \prod_{i=1}^d \llbracket a_i^{(j)}, b_i^{(j)} \rrbracket$ with update values C_j for $1 \leq j \leq L$

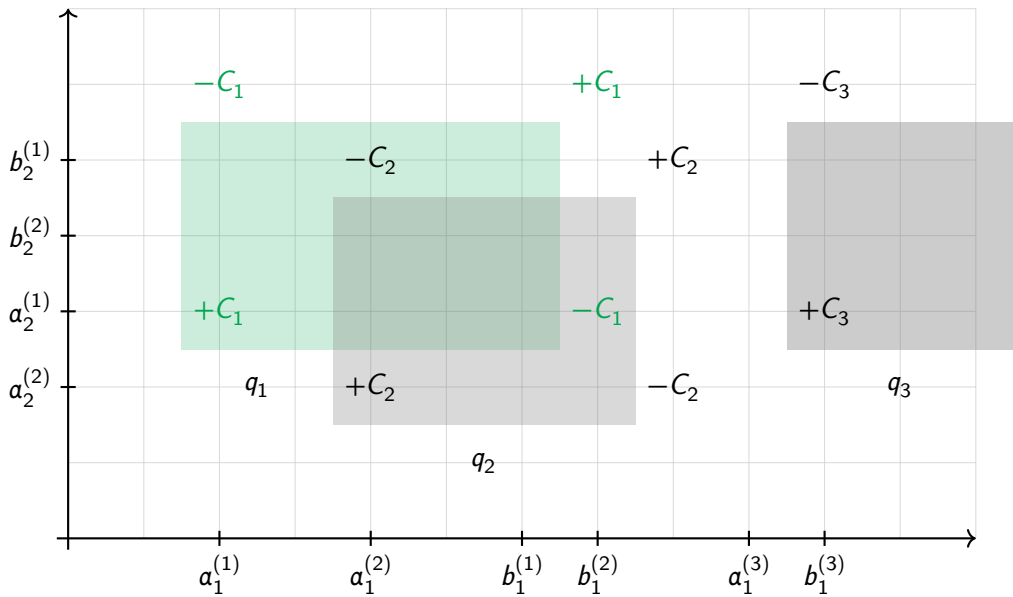
Output: The updated function $F = f + \sum_{i=1}^L q_i$

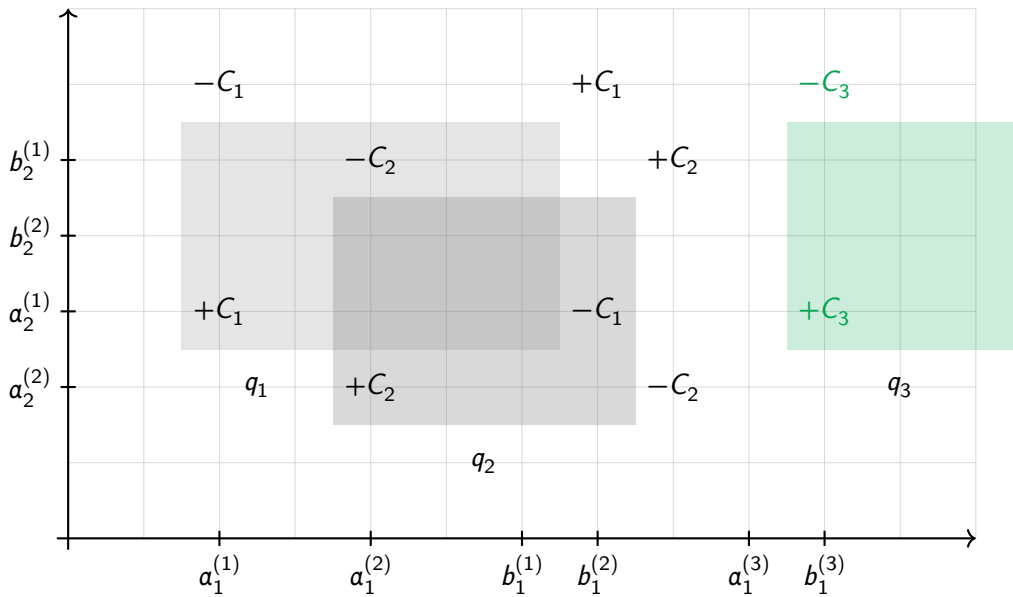
```
1:  $F \leftarrow$  The map  $f : D \rightarrow \mathbb{R}$        $\triangleright$  For an in situ implementation set  $F = f$  without any copy
   operations
2: for  $j \in \llbracket 1, L \rrbracket$  do
3:   for  $k \in D_j$  do
4:      $F(k) \leftarrow F(k) + C_j$ 
5:   end for
6: end for
   return  $F$ 
```

Runtime: With $N := \text{card}(D)$,
Non-situ: $\mathcal{O}(N + L \cdot N) = \mathcal{O}(L \cdot N)$
In-situ: $\mathcal{O}(L \cdot N)$









Definition (difference arrays): Let

- $D = \prod_{i=1}^d \llbracket n_i, m_i \rrbracket \subseteq \mathbb{Z}^d$ be a finite grid domain,
- $q_1, \dots, q_L$ be a series of constant update queries on D over the update ranges $D_j = \prod_{i=1}^d \llbracket a_i^{(j)}, b_i^{(j)} \rrbracket$ with update values C_j for $j \in \{1, \dots, L\}$,
- $M_j := \prod_{i=1}^d \left(\{a_i^{(j)}\} \cup \{b_i^{(j)} + 1 \mid b_i^{(j)} \neq m_i\} \right)$ for $j \in \{1, \dots, L\}$.

The *difference array* $\Delta[q_1, \dots, q_L] : D \rightarrow \mathbb{R}$ is defined as

$$\Delta[q_1, \dots, q_L] = \sum_{j=1}^L \sum_{p \in M_j} (-1)^{N_j(p)} \cdot C_j \cdot \delta_p,$$

where $N_j(p) = \text{card}\{p_i \mid p_i = b_i^{(j)} + 1\}$ for $j \in \{1, \dots, L\}$.

The notation δ_p means the Kronecker-delta.

Procedure: DIFFARR($D, q_1, \dots, q_L$)

Input: A finite grid domain $D = \prod_{i=1}^d \llbracket n_i, m_i \rrbracket \subseteq \mathbb{Z}^d$ and a series of constant update queries $q_1, \dots, q_L$ on D over the update ranges $D_j = \prod_{i=1}^d \llbracket a_i^{(j)}, b_i^{(j)} \rrbracket$ with update values C_j for $1 \leq j \leq L$

Output: The difference array $\Delta[q_1, \dots, q_L]$

```
1:  $d \leftarrow$  Zero map  $d : D \rightarrow \mathbb{R}$ 
2: for  $j \in \{1, \dots, L\}$  do
3:    $M_j \leftarrow \prod_{i=1}^d \left( \{a_i^{(j)}\} \cup \{b_i^{(j)} + 1 \mid b_i^{(j)} \neq m_i\} \right)$ 
4:   for  $p \in M_j$  do
5:      $N(p) \leftarrow \text{card}\{p_i \mid p_i = b_i^{(j)} + 1\}$ 
6:      $d(p) \leftarrow d(p) + (-1)^{N(p)} \cdot C_j$ 
7:   end for
8: end for
9: return  $d$ 
```

Runtime: The runtime of the body of the outer for-loop is *constant*!
 $\Rightarrow$ Total runtime: $\mathcal{O}(N + L)$ with $N := \text{card}(D)$

Towards Discrete Calculus

How does this formal notion of a difference array help us in computing the updated map

$$F = f + \sum_{i=1}^L q_i \text{ more efficiently?}$$

From the 1D/2D heuristic: Cumulatively summing over $\Delta[q_1, \dots, q_L]$ should yield the total update $\sum_{i=1}^L q_i$.

Roadmap: How does Discrete Calculus come into play?

- We will see that a difference array $\Delta[q_1, \dots, q_L]$ is the sum of discrete derivatives of the queries $q_1, \dots, q_L$, that is

$$\Delta[q_1, \dots, q_L] = \sum_{i=0}^L \Delta q_i.$$

- Cumulatively summing over $\Delta[q_1, \dots, q_L]$ will be the discrete analogon of integration and thus will be the inverse operation to discrete differentiation, that is

$$\Sigma(\Delta[q_1, \dots, q_L]) = \sum_{i=0}^L \Sigma(\Delta q_i) = \sum_{i=0}^L q_i.$$

Foundations of Discrete Calculus

Definition: The i -th backward difference Δ_i is the linear operator assigning to every discrete map $f : D \subseteq \mathbb{Z}^d \rightarrow \mathbb{R}$ with $D = \prod_{i=1}^d \llbracket n_i, m_i \rrbracket \subseteq \mathbb{Z}^d$ the map $\Delta_i f : D \rightarrow \mathbb{R}$ given by

$$\begin{aligned}\Delta_i f(k_1, \dots, k_d) &:= f(k_1, \dots, k_{i-1}, k_i, k_{i+1}, \dots, k_d) \\ &\quad - f(k_1, \dots, k_{i-1}, k_i - 1, k_{i+1}, \dots, k_d)\end{aligned}$$

for all $(k_1, \dots, k_d) \in D$ with $k_i > n_i$ and

$$\Delta_i f(k_1, \dots, k_d) := f(k_1, \dots, k_{i-1}, k_i, k_{i+1}, \dots, k_d)$$

for all $(k_1, \dots, k_d) \in D$ with $k_i = n_i$.

Definition (continued): For shorthand notation, we use

$$\Delta := \Delta_1 \circ \dots \circ \Delta_d$$

to denote the *mixed backward difference* Δ .

Definition: The *i*-th prefix sum Σ_i is the linear operator assigning to every discrete function $f : D \subseteq \mathbb{Z}^d \rightarrow \mathbb{R}$ with $D = \prod_{j=1}^d \llbracket n_j, m_j \rrbracket$ the function $\Sigma_i f : D \rightarrow \mathbb{R}$ given by

$$\Sigma_i f(k_1, \dots, k_d) := \sum_{t=n_i}^{k_i} f(k_1, \dots, k_{i-1}, t, k_{i+1}, \dots, k_d)$$

for all $(k_1, \dots, k_d) \in D$. Again for shorthand notation, we use

$$\Sigma := \Sigma_1 \circ \dots \circ \Sigma_d$$

to denote the *mixed prefix sum* Σ .

Properties of backward differences and prefix sums

- It is not hard to show that backward differences and prefix sums commute, that is for every i and j we have

$$\Sigma_i \circ \Delta_j = \Delta_j \circ \Sigma_i.$$

- In particular, $\Sigma \circ \Delta = (\Sigma_1 \circ \Delta_1) \circ \dots \circ (\Sigma_d \circ \Delta_d) = \Delta \circ \Sigma$.
- The i -th backward difference Δ_i and the i -th prefix sum Σ_i are inverse to each other for all i , that is for every discrete map f we have

$$\Delta_i \circ \Sigma_i(f) = f.$$

- Prefix sums can be computed efficiently:

$$\begin{aligned} & \Sigma_i(f)(k_1, \dots, k_d) \\ &= f(k_1, \dots, k_{i-1}, k_i, k_{i+1}, \dots, k_d) + \sum_{t=n_i}^{k_i-1} f(k_1, \dots, k_{i-1}, t, k_{i+1}, \dots, k_d) \\ &= f(k_1, \dots, k_{i-1}, k_i, k_{i+1}, \dots, k_d) + \Sigma_i(f)(k_1, \dots, k_{i-1}, k_i - 1, k_{i+1}, \dots, k_d). \end{aligned}$$

Procedure: PREFIXSUM(f, j)

Input: A discrete map $f : D \rightarrow \mathbb{R}$ with $D = \prod_{i=1}^d \llbracket n_i, m_i \rrbracket \subseteq \mathbb{Z}^d$ and an index $j \in \{1, \dots, d\}$.

Output: The j -th prefix sum $\Sigma_j(f) : D \rightarrow \mathbb{R}$

```
1:  $S \leftarrow$  The discrete map  $f : D \rightarrow \mathbb{R}$ 
2: for  $(k_1, \dots, k_{j-1}, k_{j+1}, \dots, k_d) \in \prod_{\substack{i=1 \\ i \neq j}}^d \llbracket n_i, m_i \rrbracket$  do
3:   for  $t \in \llbracket n_j + 1, m_j \rrbracket$  do
4:      $S(k_1, \dots, k_{j-1}, t, k_{j+1}, \dots, k_d) \leftarrow S(k_1, \dots, k_{j-1}, t-1, k_{j+1}, \dots, k_d)$ 
        $+ S(k_1, \dots, k_{j-1}, t, k_{j+1}, \dots, k_d)$ 
5:   end for
6: end for
7: return  $S$ 
```

Runtime: The runtime of the outer for-loop lies in $\mathcal{O}\left(\left|\prod_{\substack{i=1 \\ i \neq j}}^d \llbracket n_i, m_i \rrbracket\right| \cdot |\llbracket n_j + 1, m_j \rrbracket|\right) = \mathcal{O}(|D|)$
 $\implies$ Total runtime: $\mathcal{O}(N)$ with $N = \text{card}(D)$.

Procedure: MIXEDPREFIXSUM(f)

Input: A discrete map $f : D \rightarrow \mathbb{R}$ with $D = \prod_{i=1}^d \llbracket n_i, m_i \rrbracket \subseteq \mathbb{Z}^d$

Output: The mixed prefix sum $\Sigma(f) : D \rightarrow \mathbb{R}$

```
1:  $S \leftarrow$  The discrete map  $f : D \rightarrow \mathbb{R}$ 
2: for  $j \in \{1, \dots, d\}$  do
3:    $S \leftarrow$  PREFIXSUM( $S, j$ )
4: end for
5: return  $S$ 
```

Runtime: $\mathcal{O}(N)$

Indicator functions and Heaviside step functions

Definition: Let $p = (p_1, \dots, p_d) \in \mathbb{Z}^d$. The *Heaviside step function* $H_p : \mathbb{Z}^d \rightarrow \mathbb{R}$ is the indicator function given by

$$H_p(k_1, \dots, k_d) := \begin{cases} 1, & \text{if } \forall 1 \leq i \leq d : k_i \geq p_i, \\ 0, & \text{else} \end{cases}$$

for $(k_1, \dots, k_d) \in \mathbb{Z}^d$.

Heaviside step functions are specific indicator functions with very simple mixed backward differences.

Proposition: Let $D = \prod_{i=1}^d \llbracket n_i, m_i \rrbracket \subseteq \mathbb{Z}^d$ be a finite grid domain and $p = (p_1, \dots, p_d) \in D$. The mixed backward difference ΔH_p of the Heaviside step function $H_p : D \rightarrow \mathbb{R}$ is given by

$$\Delta H_p(k_1, \dots, k_d) = \delta_p := \begin{cases} 1, & \text{if } \forall 1 \leq i \leq d : k_i = p_i, \\ 0, & \text{else.} \end{cases}$$

Proof. ■ It is straight forward to see that $H_p = \prod_{i=1}^d H_{p_i}$

- One can show that for such products we have $\Delta H_p = \prod_{i=1}^d \Delta H_{p_i}$
- Show $\Delta H_{p_i} = \delta_{p_i}$ by a simple case distinction
- Finally, $\Delta H_p = \prod_{i=1}^d \delta_{p_i} = \delta_{(p_1, \dots, p_d)}$

□

Proposition: Let $D = \prod_{i=1}^d \llbracket a_i, b_i \rrbracket \subseteq \mathbb{Z}^d$ be a finite grid domain. Then, the indicator function $\mathbb{1}_D : \mathbb{Z}^d \rightarrow \mathbb{R}$ can be written as

$$\mathbb{1}_D = \sum_{p \in \prod_{i=1}^d \{a_i, b_i+1\}} (-1)^{N(p)} H_p,$$

where $N(p) = \text{card}\{p_i \mid p_i = b_i + 1\}$.

Proof. ■ It is not hard to see that $\mathbb{1}_D(k_1, \dots, k_d) = \prod_{i=1}^d \mathbb{1}_{\llbracket a_i, b_i \rrbracket}(k_i)$

- By a simple case distinction we see that $\mathbb{1}_{\llbracket a_i, b_i \rrbracket}(k_i) = H_{a_i}(k_i) - H_{b_i+1}(k_i)$
- Hence, $\mathbb{1}_D(k_1, \dots, k_d) = \prod_{i=1}^d (H_{a_i}(k_i) - H_{b_i+1}(k_i))$
- Using the general distribution formula for products yields

$$\prod_{i=1}^d (H_{a_i}(k_i) - H_{b_i+1}(k_i)) = \sum_{J \subseteq I} (-1)^{\text{card}(J)} \left(\prod_{i \in I \setminus J} H_{a_i}(k_i) \right) \left(\prod_{j \in J} H_{b_j+1}(k_j) \right),$$

where $I = \{1, \dots, d\}$.

Proof (continued). ■ Then for every $J \subseteq I$ we have

$$\left(\prod_{i \in I \setminus J} H_{a_i}(k_i) \right) \left(\prod_{j \in J} H_{b_j+1}(k_j) \right) = H_p(k_1, \dots, k_d)$$

with $p = (p_1, \dots, p_d)$ given by

$$p_i = \begin{cases} a_i, & \text{if } i \in I \setminus J, \\ b_i + 1, & \text{if } i \in J. \end{cases}$$

- These are exactly the points in $\prod_{i=1}^d \{a_i, b_i + 1\}$, hence it is not hard to verify that indeed

$$\begin{aligned} \sum_{J \subseteq I} (-1)^{\text{card}(J)} \left(\prod_{i \in I \setminus J} H_{a_i}(k_i) \right) \left(\prod_{j \in J} H_{b_j+1}(k_j) \right) \\ = \sum_{p \in \prod_{i=1}^d \{a_i, b_i+1\}} (-1)^{N(p)} H_p(k_1, \dots, k_d) \end{aligned}$$

with $N(p) = \text{card}\{p_i \mid p_i = b_i + 1\}$.

□

Corollary: Let $D = \prod_{i=1}^d \llbracket n_i, m_i \rrbracket \subseteq \mathbb{Z}^d$ be a finite grid domain and let $q : D \rightarrow \mathbb{R}$ be a constant update query on D with update range $D_q = \prod_{i=1}^d \llbracket a_i, b_i \rrbracket$ and update value $C \in \mathbb{R}$. Then,

$$q = \sum_{p \in M} (-1)^{N(p)} \cdot C \cdot H_p,$$

where $M = \prod_{i=1}^d (\{a_i\} \cup \{b_i + 1 \mid b_i \neq m_i\})$ and $N(p) = \text{card}\{p_i \mid p_i = b_i + 1\}$. In particular, the mixed backward difference Δq of q can be expressed as

$$\Delta q = \sum_{p \in M} (-1)^{N(p)} \cdot C \cdot \delta_p.$$

Proof. ■ By definition $q = C \cdot \mathbb{1}_{D_q}$

■ By the previous proposition:

$$q = \sum_{p \in \prod_{i=1}^d \{a_i, b_i+1\}} (-1)^{N(p)} \cdot C \cdot H_p.$$

Proof (continued). ■ We are not viewing q as a function over $\mathbb{Z}^d$ but over

$$D = \prod_{i=1}^d \llbracket n_i, m_i \rrbracket \subseteq \mathbb{Z}^d$$

- If $p \in \prod_{i=1}^d \{a_i, b_i + 1\}$ with $p_j = b_j + 1$ and $b_j = m_j$, then $p \notin D$
- In this case, $H_p(k) = 0$ for all $k \in D$
- Hence, the only points $p \in \prod_{i=1}^d \{a_i, b_i + 1\}$ that matter are those with $p_j \neq b_j + 1$ if $b_j = m_j$, that is the points in

$$M = \prod_{i=1}^d (\{a_i\} \cup \{b_i + 1 \mid b_i \neq m_i\})$$

- Therefore,

$$q = \sum_{p \in M} (-1)^{N(p)} \cdot C \cdot H_p.$$

- Lastly, the form of Δq follows directly by the linearity of the mixed backward difference Δ and $\Delta H_p = \delta_p$. □

Recall the formal definition of difference arrays:

Definition: Let $D = \prod_{i=1}^d \llbracket n_i, m_i \rrbracket \subseteq \mathbb{Z}^d$ be a finite grid domain and $q_1, \dots, q_L$ a series of constant update queries on D . The *difference array* $\Delta[q_1, \dots, q_L] : D \rightarrow \mathbb{R}$ is defined as

$$\Delta[q_1, \dots, q_L] = \sum_{j=1}^L \sum_{p \in M_j} (-1)^{N_j(p)} \cdot C_j \cdot \delta_p,$$

where $M_j = \prod_{i=1}^d \left(\{a_i^{(j)}\} \cup \{b_i^{(j)} + 1 \mid b_i^{(j)} \neq m_i\} \right)$ and $N_j(p) = \text{card}\{p_i \mid p_i = b_i^{(j)} + 1\}$ for $1 \leq j \leq L$.

- With the corollary on the last slide we see that

$$\Delta q_j = \sum_{p \in M_j} (-1)^{N_j(p)} \cdot C_j \cdot \delta_p$$

for all $j \in \{1, \dots, L\}$.

- Hence, we indeed have

$$\Delta[q_1, \dots, q_L] = \sum_{j=1}^L \Delta q_j.$$

- The prefix sum and the mixed backwards difference are inverse to each other, thus

$$\Sigma(\Delta[q_1, \dots, q_L]) = \sum_{j=1}^L \Sigma(\Delta q_j) = \sum_{j=1}^L q_j.$$

- We have seen that $\Delta[q_1, \dots, q_L]$ can be computed in $\mathcal{O}(L + N)$.
- We have seen that the prefix sum $\Sigma(\Delta[q_1, \dots, q_L])$ can be computed in $\mathcal{O}(N)$.
- The addition $f + \Sigma(\Delta[q_1, \dots, q_L])$ has time complexity $\mathcal{O}(N)$.
- In total: Computing $f + \Sigma(\Delta[q_1, \dots, q_L])$ can be done in $\mathcal{O}(L + N)$.

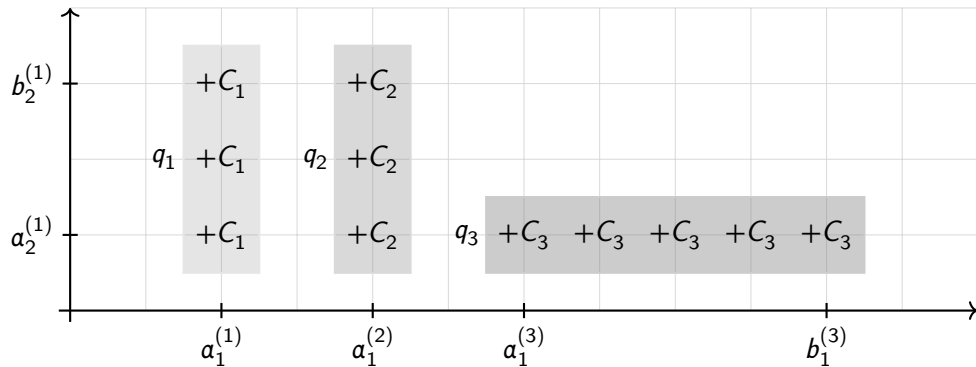
Input: A discrete map $f : D \rightarrow \mathbb{R}$ with $D = \prod_{i=1}^d \llbracket n_i, m_i \rrbracket$ and a series of constant update queries $q_1, \dots, q_Q$ on D

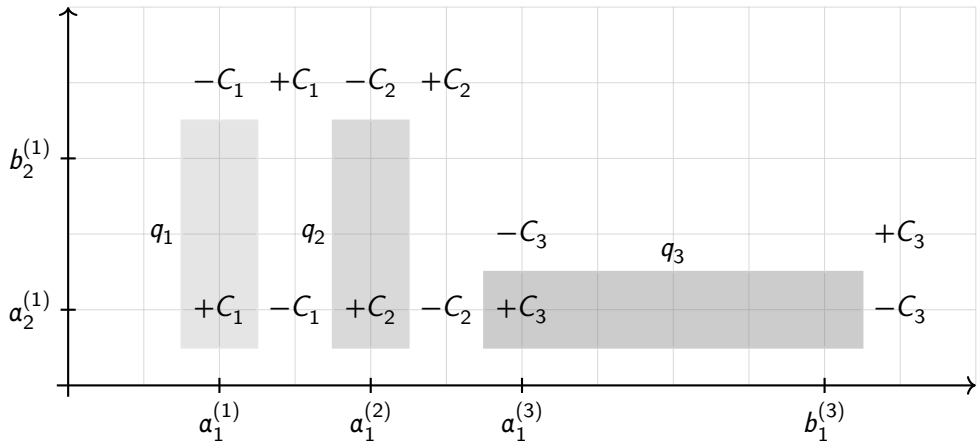
Output: The updated map $F = f + \sum_{i=1}^Q q_i$

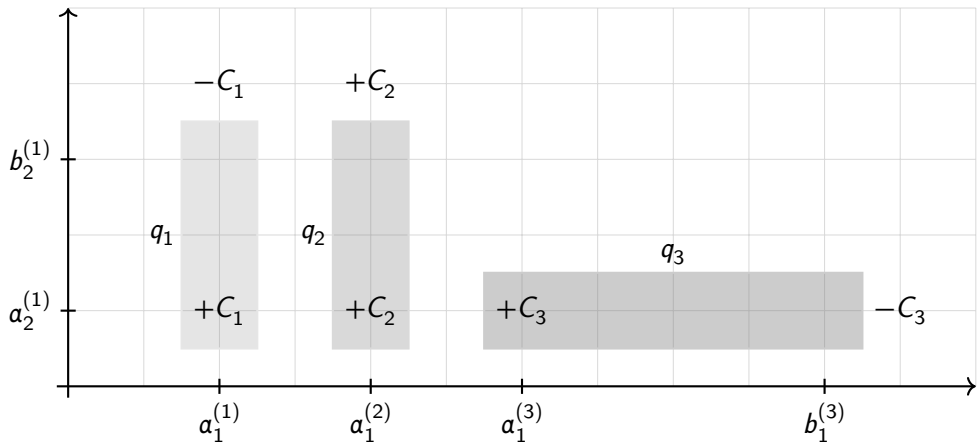
- 1: $d \leftarrow \text{DIFFARR}(D, q_1, \dots, q_Q)$
- 2: $\Sigma d \leftarrow \text{MIXEDPREFIXSUM}(d)$
- 3: $F \leftarrow f + \Sigma d$
- 4: **return** F

Runtime: $\mathcal{O}(L+N+N+N) = \mathcal{O}(L+N)$

Flat update queries







Consecutive update queries

